

Question number	Answer	Additional guidance	Mark
7(a)(i)	B L-DOPA crosses the blood brain barrier and is then converted to dopamine in the brain		Computer (1)

Question number	Answer	Additional guidance	Mark
7(a)(ii)	An answer that includes three of the following points: • fewer dopamine molecules to bind to the {ligand gated sodium channels/receptors} (1) • on the post-synaptic membrane (1) • initiating fewer action potentials in (post-synaptic neurone) (1) • fewer impulses sent to parts of brain controlling {motor function / muscles} (1)	ALLOW fewer impulses sent to muscles / effectors	Expert (3)

Question number	Answer	Additional guidance	Mark
7(b)	An answer that includes the following points: • {reduced / stop} influx of calcium ions into pre-synaptic neurone (1) • No fusion of (secretory) vesicles with the pre-synaptic membrane (1) or • changes to {pre-synaptic / vesicle} membrane proteins (1) • prevents fusion of vesicles with the pre-synaptic membrane (1)	ALLOW fewer vesicles fuse with pre-synaptic membrane.	Expert (2)

Question number	Answer	Additional guidance	Mark
7(c)	A description that includes the following points: • microarrays allow identification of {active genes / gene transcription} (1) • the activity of many genes can be analysed in a single sample (1) • by collecting information about genetic differences from many individuals (with or without Parkinson's) (1) • bioinformatics /computers/databases /algorithms used to analyse the data (1) • (key) differences between healthy and Parkinson's disease	ALLOW Develop algorithms to identify genomes / gene sequences	Expert (4)